

Air Quality Health Index (AQHI) — Edmonton

3.0

Current AQHI at Edmonton Lendrum (18.14 km from venue)
Forecast Sun 9:00 PM at Edmonton Lendrum (18.14 km from venue): **2.2**

Health Risk	AQHI	At-Risk Population	General Population
Low Risk	3.0	Enjoy your usual outdoor activities.	Ideal air quality for outdoor activities.

No active Environment Canada weather alerts for the venue.

Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 18.2°C and feels near 17.0°C. Winds are 20 km/h from NW, gusting near 37 km/h.
- The nearest current AQHI is 3 at Edmonton Lendrum, 18.1 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 3 (forecast tonight: 5, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 6.1 km away) puts the area near 3.
- The nearest air monitoring station (Leduc Sensor, 12.7 km) reports fine particulate matter (PM2.5) at 25.9 µg/m³.
- A nearby community sensor ('ACA_BeaumontCityHall', 1.5 km) reads 14.9 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 117.0 km to the W, i.e. arriving from the W.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-23) is 135.6 km SW of the venue.
- The AQHI forecast for Sun 9:00 PM is 2, based on Edmonton Lendrum, 18.1 km from the venue.
- Thunderstorm conditions appear in the hourly forecast beginning around Sun 8:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are precipitation, thunderstorm.

Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

Hazard Snapshot — Overall Risk: High Risk

AIR QUALITY Low Risk 3.0 AQHI	HEAT Low Risk 22.2 °C apparent/humidex	WIND Low Risk 41.8 km/h peak gust	PRECIPITATION High Risk 3.4 mm/h maximum	THUNDERSTORM High Risk Sun 8:00 PM first forecast signal
AQHI RATE OF CHANGE Low Risk 0.0 AQHI change/hour			WIND SHEAR Low Risk 2.4 ° direction diff (10m vs 120m)	

Local Monitoring

Nearest station 3.0 Edmonton Lendrum · 18.14 km	Blended estimate 3.0 confidence High	PM2.5 (station) 25.9 µg/m³ Leduc Sensor · 12.73 km	PM2.5 (community) 14.9 µg/m³ ACA_BeaumontCityHall · 1.54 km
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